WORKED SOLUTION

STEP II 2013, Question 13

Probability and Statistics • Runs in Coin Tossing

STEP II 2013, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q13

The key idea

The event $S = n$ says exactly what the first $n+1$ tosses are: n equal tosses, then one that breaks the run. The same is true of $A = n$: n alternating tosses, then a repeat. So each probability is a sum of two products, one for each possible first toss. Comparing S with A then comes down to factorising a difference of four terms, and a factorisation is easier to see if we keep p and q as independent letters rather than replacing q by $1-p$.

Solution

Throughout, tosses are independent, each a head with probability p and a tail with probability q, and $p, q > 0$ with $p \neq q$.

The general formulae

$S = n$ happens exactly when tosses $1, \ldots, n$ all match the first and toss $n+1$ is different. If the first toss is a head this is $\mathrm{H}^n\mathrm{T}$, with probability $p^n q$; if it is a tail it is $\mathrm{T}^n\mathrm{H}$, with probability $q^n p$. These cases are disjoint, so

$$\mathrm{P}(S = n) = p^n q + q^n p = pq\left(p^{n-1} + q^{n-1}\right).$$

$A = n$ happens exactly when tosses $1, \ldots, n$ alternate and toss $n+1$ is the *same* as toss n, which ends the alternation.

- For $n = 2m$ (even): the run $\mathrm{HTHT}\cdots\mathrm{HT}$ has m heads and m tails and ends in T, so the next toss must be T, giving $p^m q^m \cdot q$. Starting with a tail gives $q^m p^m \cdot p$. Hence
$$\mathrm{P}(A = 2m) = p^m q^m (p+q).$$
- For $n = 2m+1$ (odd): the run $\mathrm{HTH}\cdots\mathrm{H}$ has $m+1$ heads and m tails and ends in H, so the next toss must be H, giving $p^{m+2}q^m$. Starting with a tail gives $q^{m+2}p^m$. Hence
$$\mathrm{P}(A = 2m+1) = p^m q^m\left(p^2 + q^2\right).$$

Part (i)

$A = 1$ means the second toss repeats the first: HH or TT. So $\mathrm{P}(A = 1) = p^2 + q^2$ (the case $m = 0$ above). $S = 1$ means the second toss differs from the first: HT or TH, so

$$\mathrm{P}(S = 1) = pq + qp = 2pq.$$

Then

$$\mathrm{P}(A = 1) - \mathrm{P}(S = 1) = p^2 - 2pq + q^2 = (p - q)^2 > 0,$$

strictly because $p \neq q$. So $\mathrm{P}(S = 1) < \mathrm{P}(A = 1)$.

Part (ii)

From the formulae,

$$\mathrm{P}(S = 2) = p^2 q + q^2 p = pq(p + q) = \mathrm{P}(A = 2),$$

as required. For length 3,

$$\mathrm{P}(S = 3) = p^3 q + q^3 p = pq\left(p^2 + q^2\right) = \mathrm{P}(A = 3),$$

so these two are equal as well.

Part (iii)

Even lengths. Take $n > 1$. Dividing both probabilities by $pq > 0$,

$$\frac{\mathrm{P}(S = 2n)}{pq} = p^{2n-1} + q^{2n-1}, \qquad \frac{\mathrm{P}(A = 2n)}{pq} = p^{n-1}q^{n-1}(p + q) = p^n q^{n-1} + p^{n-1}q^n.$$

? **Ask yourself** How can we tell the sign of $p^{2n-1} + q^{2n-1} - p^n q^{n-1} - p^{n-1}q^n$?

Pair each pure power with the cross term that shares its highest power of the same letter:

$$\begin{aligned} p^{2n-1} + q^{2n-1} - p^n q^{n-1} - p^{n-1}q^n &= p^n\left(p^{n-1} - q^{n-1}\right) - q^n\left(p^{n-1} - q^{n-1}\right) \\ &= \left(p^n - q^n\right)\left(p^{n-1} - q^{n-1}\right). \end{aligned}$$

Since $p, q > 0$ and $p \neq q$, for any $k \geq 1$ the number $p^k - q^k$ is non-zero and has the same sign as $p - q$ (as $x \mapsto x^k$ is strictly increasing on $x > 0$). Here $n \geq 2$, so both n and $n - 1$ are at least 1, and the two brackets are non-zero with the same sign. Their product is positive, and multiplying back by $pq > 0$,

$$\mathrm{P}(S = 2n) > \mathrm{P}(A = 2n) \qquad (n > 1).$$

(For $n = 1$ the second bracket is $p^0 - q^0 = 0$, which is the equality of part (ii).)

Odd lengths. In the same way, again with $n > 1$,

$$\frac{\mathrm{P}(S = 2n+1)}{pq} = p^{2n} + q^{2n},$$

$$\frac{\mathrm{P}(A = 2n+1)}{pq} = p^{n-1}q^{n-1}\left(p^2 + q^2\right) = p^{n+1}q^{n-1} + p^{n-1}q^{n+1},$$

and

$$\begin{aligned} p^{2n} + q^{2n} - p^{n+1}q^{n-1} - p^{n-1}q^{n+1} &= p^{n+1}\left(p^{n-1} - q^{n-1}\right) - q^{n+1}\left(p^{n-1} - q^{n-1}\right) \\ &= \left(p^{n+1} - q^{n+1}\right)\left(p^{n-1} - q^{n-1}\right). \end{aligned}$$

By the same argument both brackets are non-zero with the sign of $p - q$, so

$$\mathrm{P}(S = 2n+1) > \mathrm{P}(A = 2n+1) \qquad (n > 1).$$

Again the factor $p^{n-1} - q^{n-1}$ vanishes at $n = 1$, which matches $\mathrm{P}(S = 3) = \mathrm{P}(A = 3)$.

So, in summary: a straight run of length 1 is less likely than an alternating one, lengths 2 and 3 are equally likely, and every length from 4 upwards is more likely as a straight run.

Checks

- Take $p = \frac{1}{3}$, $q = \frac{2}{3}$. Then $\mathrm{P}(S = 4) = pq(p^3 + q^3) = \frac{2}{9} \cdot \frac{1}{3} = \frac{2}{27} \approx 0.074$ and $\mathrm{P}(A = 4) = p^2q^2 = \frac{4}{81} \approx 0.049$, so $S = 4$ is indeed more likely.
- Summing $\mathrm{P}(S = n)$ over all $n \geq 1$ gives $q \cdot \frac{p}{1-p} + p \cdot \frac{q}{1-q} = p + q = 1$, as it should.
- Each formula was confirmed by computer, enumerating all sequences of 14 tosses with exact fractions for several values of p, for lengths up to 13.

Common slips

- Forgetting the toss that *ends* the run. $S = n$ needs toss $n+1$ to differ; without it you are computing $\mathrm{P}(S \geq n)$.
- Ending an alternating run on the wrong letter: a run of odd length ends on the same letter it started with, and one of even length ends on the other letter.
- Substituting $q = 1 - p$ in part (iii): the expressions become polynomials in p whose factorisation is much harder to spot.

Beyond STEP

S is a geometric-type random variable: given the first toss, the run length is the waiting time for the first change. Runs like these underlie the runs test in statistics, used to

judge whether a sequence of outcomes looks random, and generating functions give a neat way to find the full distribution of longest runs. Grimmett and Welsh, *Probability: An Introduction*, covers generating functions and waiting times at the right level.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.